

Department of Electrical, Computer and Systems Engineering

Rensselaer Polytechnic Institute

**ECSE 4560/6560 Modern Communication Systems
Fall 2025**

Midterm Exam 1

ECSE 4560: There are 4 problems +MCQ/TF Questions, worth 50 points and 11 pages in this exam booklet.

ECSE 6560: There are 5 problems +MCQ/TF Questions, worth 55 points and 11 pages in this exam booklet.

A note from the instructors:

If you find a question ambiguous, be sure to write down any assumptions you make and provide appropriate reasoning. **Be neat and legible.**

This is an **open-book, open-notes** exam with **printed/handwritten materials only**; **no electronic devices** (no laptops, tablets, phones, calculators, smartwatches) are permitted. You have a total of 80 min to complete the exam—manage your time across all problems. Write legibly, **show key steps** for partial credit, and box or clearly highlight your final answers.

Please write your name in the space below.

Name: _____

RIN: _____

True/False

[10 points]

1. In BPSK over AWGN with equal priors, the ML and MAP decision rules yield the same decision threshold. (T/F)
2. The thermal noise power at room temperature scales linearly with both absolute bandwidth and temperature. (T/F)
3. For M -QAM, SER is always strictly lower than BER at the same E_b/N_0 . (T/F)
4. Ideal bandlimited sampling at $f_s > 2W$ permits perfect reconstruction using an ideal low-pass filter. (T/F)
5. In a narrowband (flat) fading channel, the multipath channel can be well-approximated by a single complex gain. (T/F)
6. Inter-symbol interference (ISI) becomes more severe as the symbol time approaches the delay spread of the channel. (T/F)
7. For a real passband signal, the spectrum is conjugate-symmetric; for complex baseband, it need not be. (T/F)
8. Frequency-selective fading can occur even when the overall signal bandwidth is very small compared to the channel's coherence bandwidth. (T/F)
9. In OFDM, orthogonality allows tightly packed subcarriers without mutual interference at the sampling instants. (T/F)
10. Carrier frequency offset (CFO) causes a common phase rotation across subcarriers that accumulates over OFDM symbols. (T/F)

Multiple Choice Questions

[10 points]

1. In AWGN with equal priors for BPSK, the optimal decision boundary for $y = s + v$ is:
 - (a) $y = +A$
 - (b) $y = 0$
 - (c) $y = -A$
 - (d) $y = \frac{\sigma^2}{2A} \ln \frac{P(-A)}{P(+A)}$
2. With unequal priors $P(+A) \neq P(-A)$, the MAP threshold relative to 0:
 - (a) does not change
 - (b) becomes $y = A$
 - (c) shifts toward the more likely symbol
 - (d) shifts toward the less likely symbol
3. For a fixed E_b/N_0 , which has the lowest BER among these uncoded schemes?
 - (a) 64-QAM
 - (b) 16-QAM
 - (c) QPSK
 - (d) All are identical
4. Perfect reconstruction of a bandlimited signal of one-sided bandwidth W requires:
 - (a) $f_s > W$
 - (b) $f_s = 2W$
 - (c) $f_s \geq 2W$ with ideal LPF
 - (d) $f_s < 2W$ with ideal LPF
5. “Flat fading” implies:
 - (a) $H(f)$ is constant across the signal band
 - (b) $H(f)$ has deep notches across the band
 - (c) the channel has no delay
 - (d) the channel is time-varying

6. In OFDM, a sufficient condition for no ISI between symbols is:
- (a) $T_{\text{CP}} \geq T$
 - (b) $T_{\text{CP}} > \max_k \tau_k$
 - (c) $T > \max_k \tau_k$
 - (d) $F_s > 2B$
7. The main reason OFDM needs a cyclic prefix is to:
- (a) preserve orthogonality in the presence of delay spread
 - (b) detect the start of a packet
 - (c) reduce Peak-to-average-power ratio
 - (d) eliminate phase noise and CFO
8. Frequency-selective fading is more severe when:
- (a) signal bandwidth $\ll$ coherence bandwidth
 - (b) signal bandwidth $\approx$ coherence bandwidth
 - (c) signal bandwidth $\gg$ coherence bandwidth
 - (d) coherence time is large
9. CFO after coarse correction typically shows up as:
- (a) amplitude ripple across subcarriers only
 - (b) constant phase offset per subcarrier, no time accumulation
 - (c) time-accumulating common phase rotation across symbols
 - (d) frequency-dependent group delay
10. Compared to single-carrier, OFDM's key benefit in frequency-selective channels is:
- (a) removing the need for synchronization
 - (b) eliminating noise
 - (c) avoiding the need for pilots
 - (d) converting a wideband frequency-selective channel into many flat subchannels

Question 1: Modulation, Sampling and Channel Effects

- (a) **[2 points]** For a 4-QAM constellation with average symbol energy E_s , find the minimum Euclidean distance $d_{\min}$ between symbols.
- (b) **[2 points]** A baseband signal limited to $B = 3$ MHz is sampled at $f_s = 4$ MHz. Sketch (qualitatively) its sampled spectrum and identify the aliasing regions. Properly label your sketch diagram.
- (c) **[2 points]** Two propagation paths have lengths $d_1 = 10$ m and $d_2 = 10.06$ m at $f_c = 5$ GHz. Compute the phase difference between the paths and state whether the interference is constructive or destructive.

Question 2: BER of OOK

Consider on-off keying (OOK) with symbols $s \in \{0, A\}$ sent over a real AWGN channel with coherent detection at the receiver. The received sample is:

$$y = s + v, \quad v \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{N_0}{2}.$$

Assume equiprobable bits unless otherwise stated. Use the Q -function $Q(x) \triangleq \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt$ wherever helpful.

- (a) **[2 points]** State the decision boundary (threshold) that decides $\hat{s} \in \{0, A\}$ from y under equal priors. No derivation required; justify briefly in one sentence.

- (b) **[4 points]** Derive the average bit error probability P_b for equal priors in terms of A and σ using the Q -function.

(c) **[2 points]** For one bit per symbol and a unit-energy pulse shape, relate the amplitude A to the average energy per bit E_b .

(d) **[2 points]** Using $\sigma^2 = \frac{N_0}{2}$, express P_b entirely in terms of E_b/N_0 .

(e) **[2 points]** Compare the resulting OOK P_b with the coherent BPSK result $P_b^{\text{BPSK}} = Q(\sqrt{2E_b/N_0})$.

To achieve a fixed target BER, does OOK requires same SNR or lower SNR or higher SNR compared to BPSK? If different, quantify the SNR difference (in dB) to achieve the target BER and explain why in one sentence.

Question 3: Multipath Channel

Let the complex baseband transmit signal be $x_b(t)$ and its complex passband representation be

$$x_p(t) = x_b(t) e^{j2\pi f_c t}.$$

The physical multipath channel produces the passband output

$$y_p(t) = \sum_{k=1}^K h_k x_p(t - \tau_k) + v_p(t),$$

where each path has complex gain h_k and delay τ_k , and $v_p(t)$ is passband noise. Define the IQ-mixer output at the receiver as

$$y_b(t) = y_p(t) e^{-j2\pi f_c t}.$$

- (a) **[5 points]** Derive a single closed-form expression for the complex baseband received signal $y_b(t)$ in terms of $x_b(t)$, f_c , $\{h_k\}$, and $\{\tau_k\}$.

Question 4: OFDM System Design

A stadium deploys a temporary microcell for event-night connectivity. The radio team decides to use OFDM with occupied bandwidth $B = 20$ MHz and subcarrier spacing $\Delta f = 15$ kHz. To maximize data rate for a short demo, they plan to *use all subcarriers within the bandwidth for payload* (no pilots, no guard bands), and they will start with *no cyclic prefix (CP)* before later adding one to handle multipath. Modulation is fixed to QPSK across all subcarriers. Coding rate is 1.

- (a) [3 points] From the occupied bandwidth and target subcarrier spacing, compute the **needed number of subcarriers** and a convenient power-of-two **FFT size** N_{FFT} . What DAC **sampling rate** f_s is required to stream this OFDM signal?
- (b) [2 points] Compute the useful OFDM symbol duration T_u in μs for $\Delta f = 15$ kHz.
- (c) [2 points] With QPSK on all occupied subcarriers and *no CP*, compute the net physical-layer throughput in Mbit/s. (Hint: Use the count of useful QPSK symbols per second implied by T_u .)

Question 5 (ECSE 6560 Only): Residual CFO and SFO

A drone transmits live telemetry over a 5 GHz OFDM link (bandwidth $B = 20$ MHz, number of subcarriers $N = 64$, cyclic prefix $N_{CP} = 16$ samples). The system transmits 100 OFDM symbols per packet. The ground receiver uses standard preamble-based coarse CFO correction and channel estimation.

During flight tests, the engineering team notices that, after a few OFDM symbols, the constellation at the receiver begins to slowly rotate in phase with time, and shows non-uniform stretching or compression across subcarriers.

The suspicion is that the receiver suffers from **residual carrier frequency offset (CFO)** and **sampling frequency offset (SFO)** even after initial synchronization.

The team fit a linear model $\Delta\phi(f) = mf + b$ and compute the slope $m = 0.05$ rad/index and intercept $b = -0.07$ rad. Using the OFDM phase-tracking relation

$$\Delta\phi(f) = 2\pi f \frac{(N + N_{CP})}{N} \delta T_s - 2\pi \delta f_c (N + N_{CP}) T_s,$$

they derive numerical values for residual CFO $\delta f_c = 2.8$ kHz and SFO time offset $\delta T_s = 0.0064$ μ s.

- (a) **[2 points]** Estimate the cumulative constellation rotation (in degrees) caused solely by the residual CFO after 100 OFDM symbols. Comment on whether this would cause visible rotation in a constellation plot.

- (b) **[3 points]** If the system were to double its FFT size to $N = 128$ while keeping the same sampling rate, how would the impact of CFO and SFO values as identified above change the accumulated phase error per symbol? Provide a short justification based on your expression above.

End of Final Exam